

Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Recall:

pivot: first non-zero entry in a row

row echelon form?

each pivot to the right of those above?

zero rows at bottom?

reduced row echelon form?

additionally:

all pivots equal 1?

above each pivot, all zeroes in its column?

Next up:

- This week: Chapter 2.4-8 of *LADW*
- ***Change:*** Only Quiz 3 on Friday.
- Open office hours today, 3:30-4:30 in Eckhart 207A.
- Quiz 1 & 2 returned. First requizzing session on Wed (4-5pm Eck 207A)
- Homework 2 is due at the end of the week.
- ***Change:*** I won't be away on Friday, April 21.

Warm-up 1

For each of the following matrices, decide: is it in row echelon form? is it in reduced row echelon form? When the answer is yes, put a check. When the answer is no, apply row reduction to achieve these forms.

matrix:

$$\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 4 & 4 \end{pmatrix}$$

row echelon form?

reduced row echelon form?

Warm-up 1

For each of the following matrices, decide: is it in row echelon form? is it in reduced row echelon form? When the answer is yes, put a check. When the answer is no, apply row reduction to achieve these forms.

matrix:

$$\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 2 & 2 \\ 4 & 4 \end{pmatrix}$$

row echelon form?

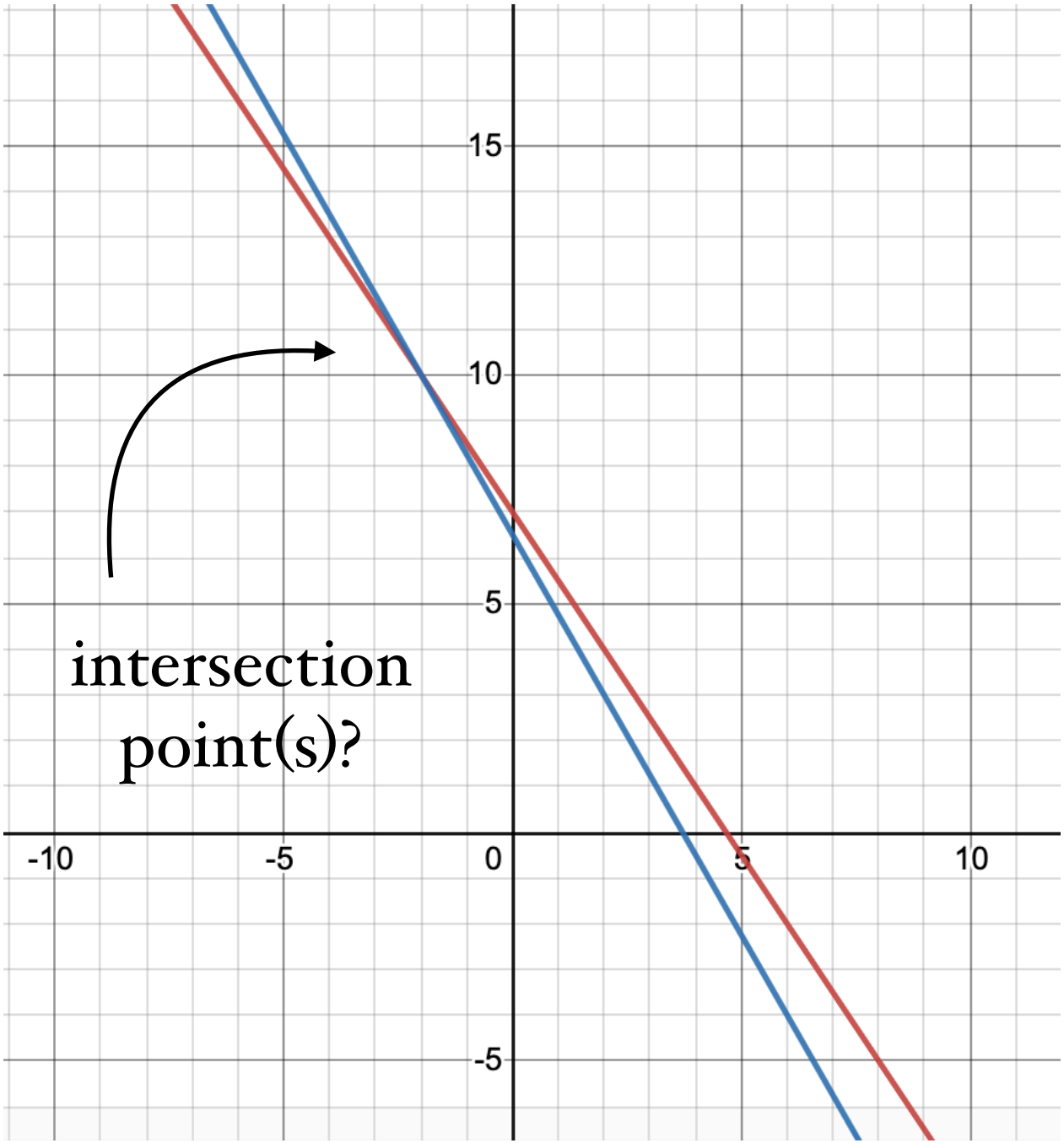


reduced row echelon form?



system of
linear equations

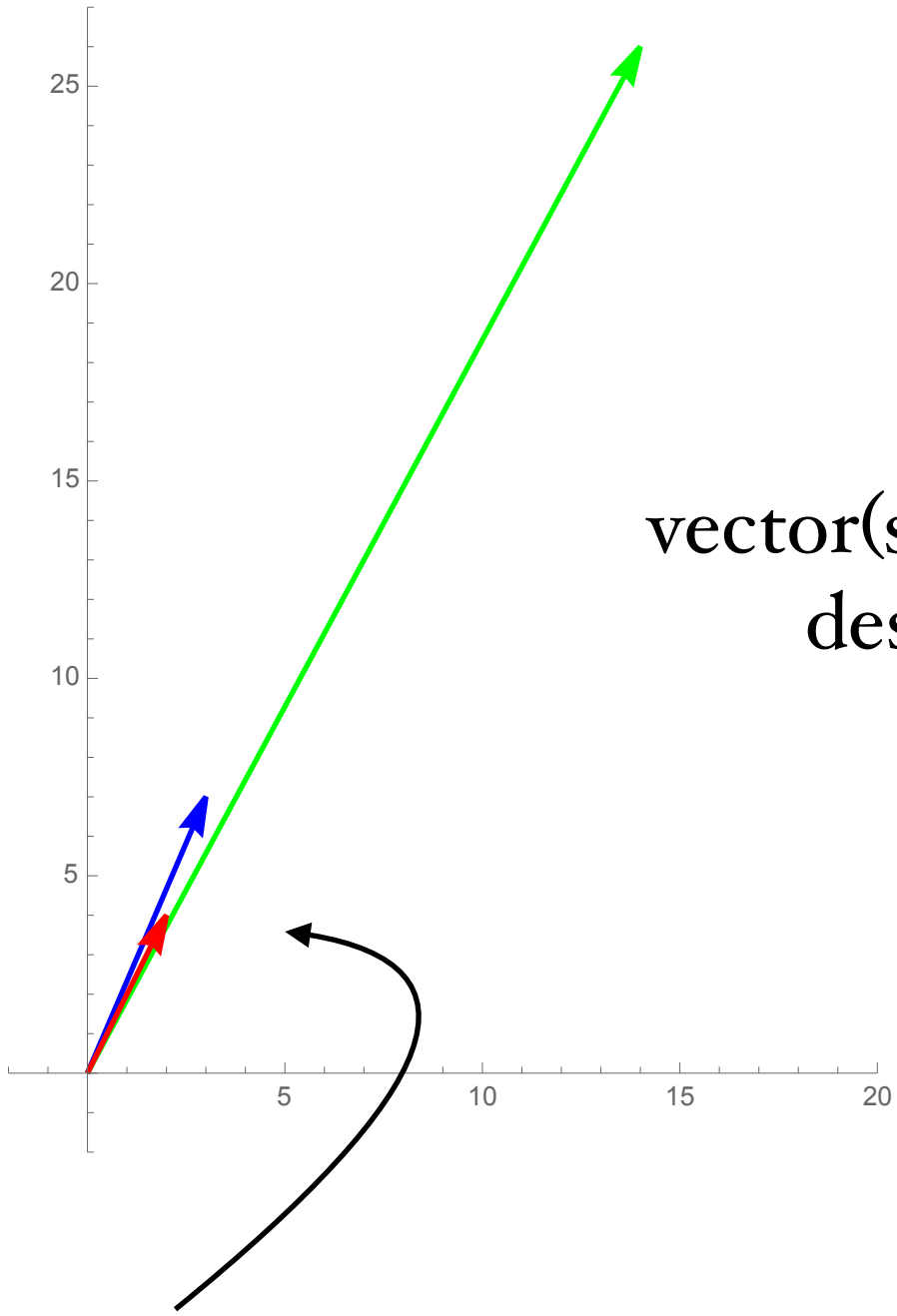
$$\begin{aligned} 3x + 2y &= 14 \\ 7x + 4y &= 26 \end{aligned}$$



$x = -2, y = 10$

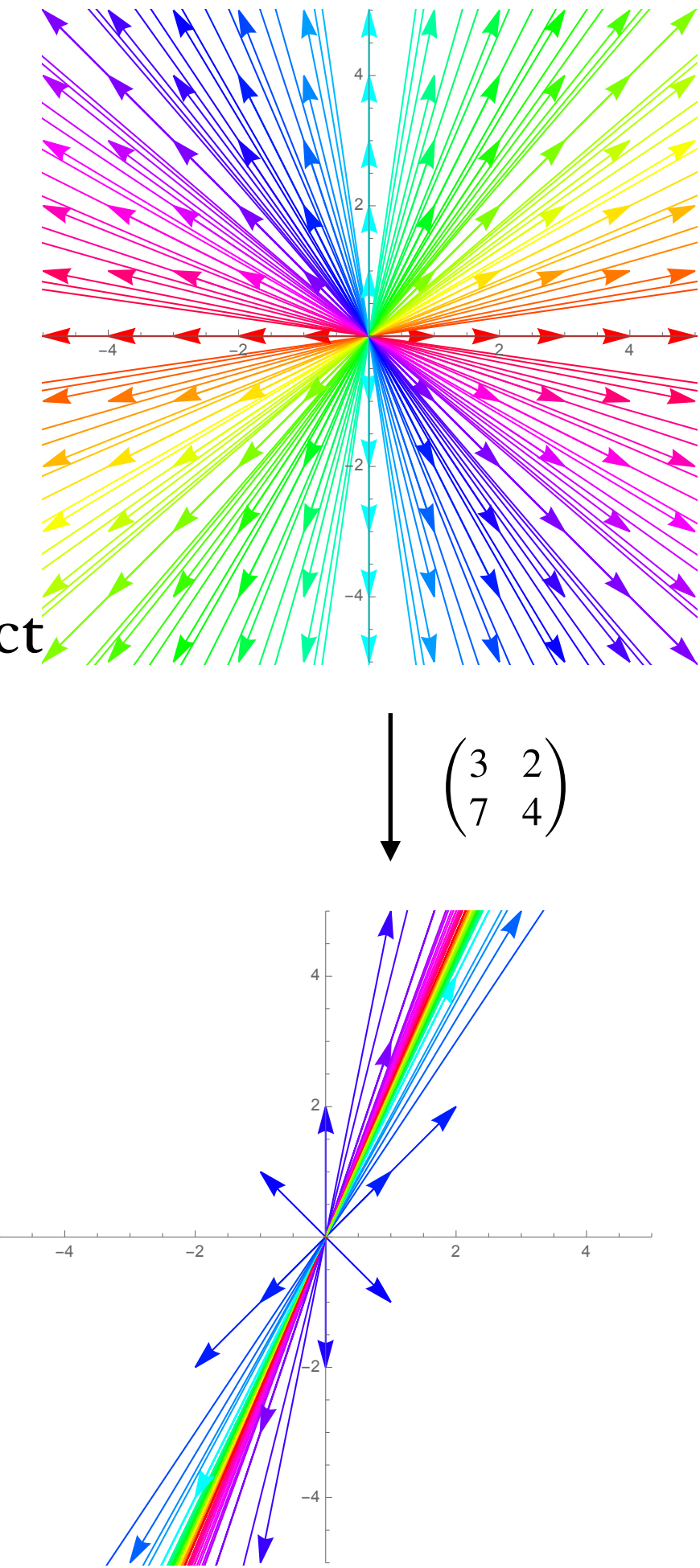
vector equation

$$x \begin{pmatrix} 3 \\ 7 \end{pmatrix} + y \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 14 \\ 26 \end{pmatrix}$$



matrix equation

$$\begin{pmatrix} 3 & 2 \\ 7 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 14 \\ 26 \end{pmatrix}$$

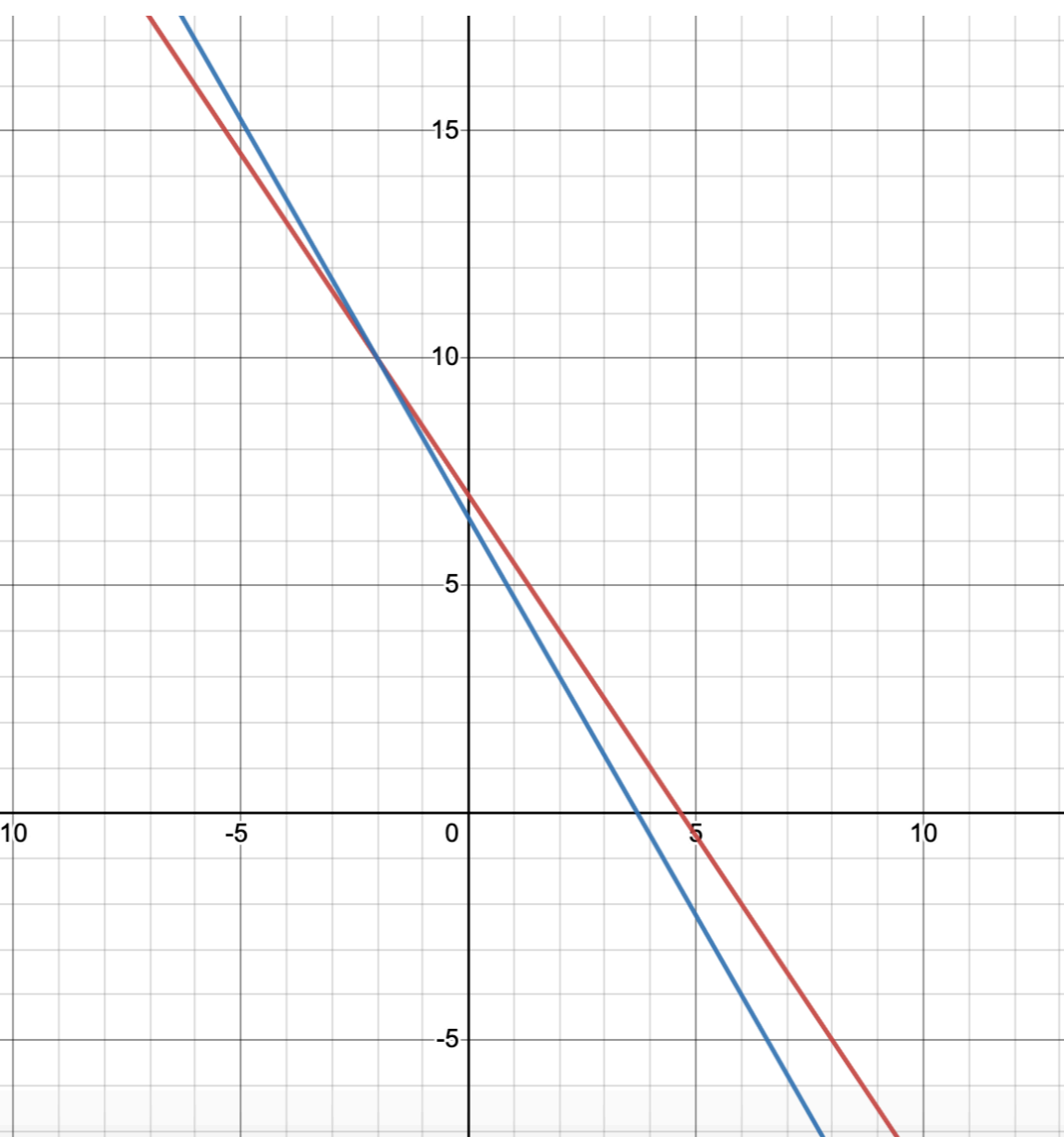


In what way does applying row operations
change these problems?

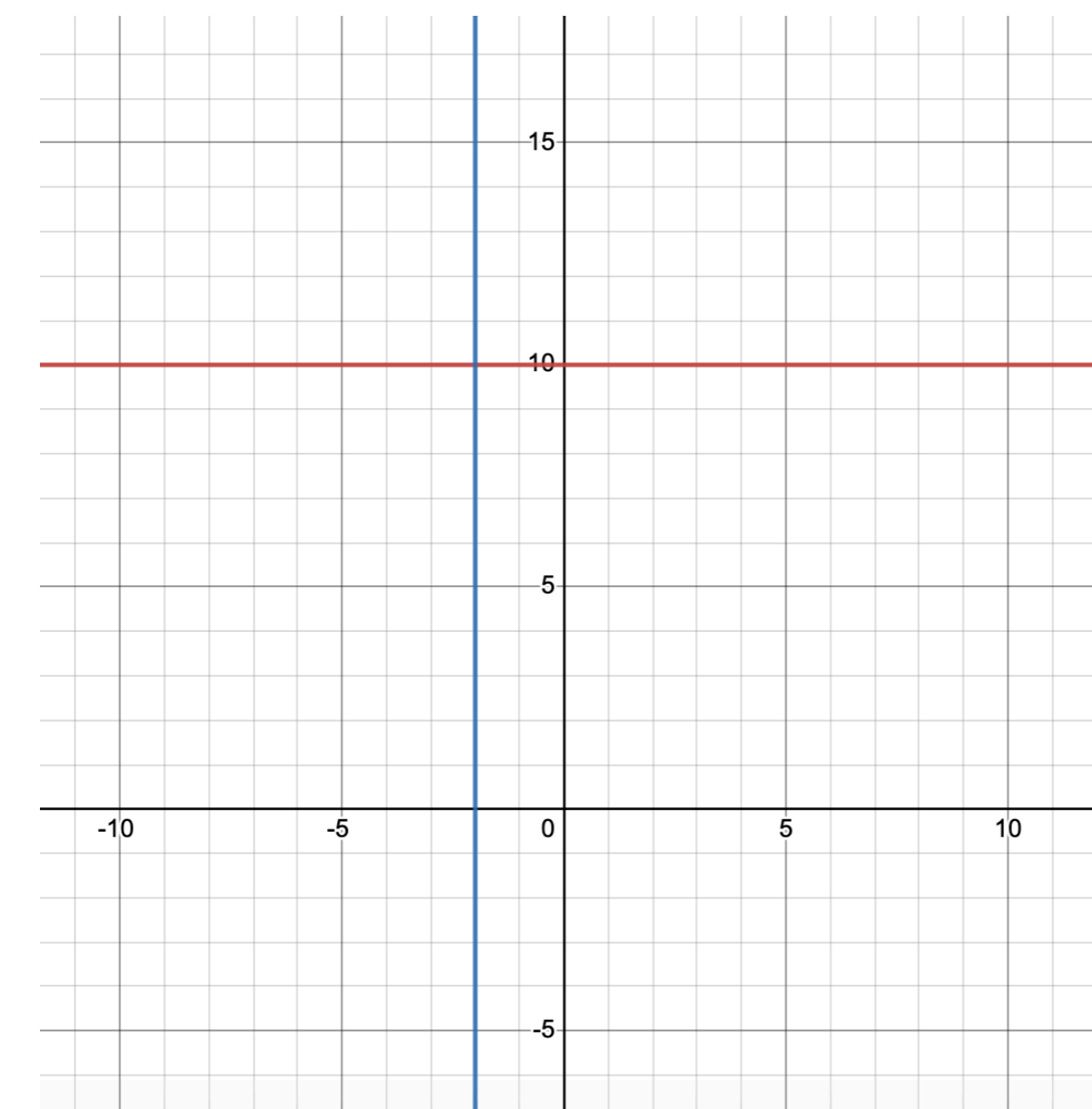
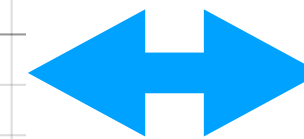
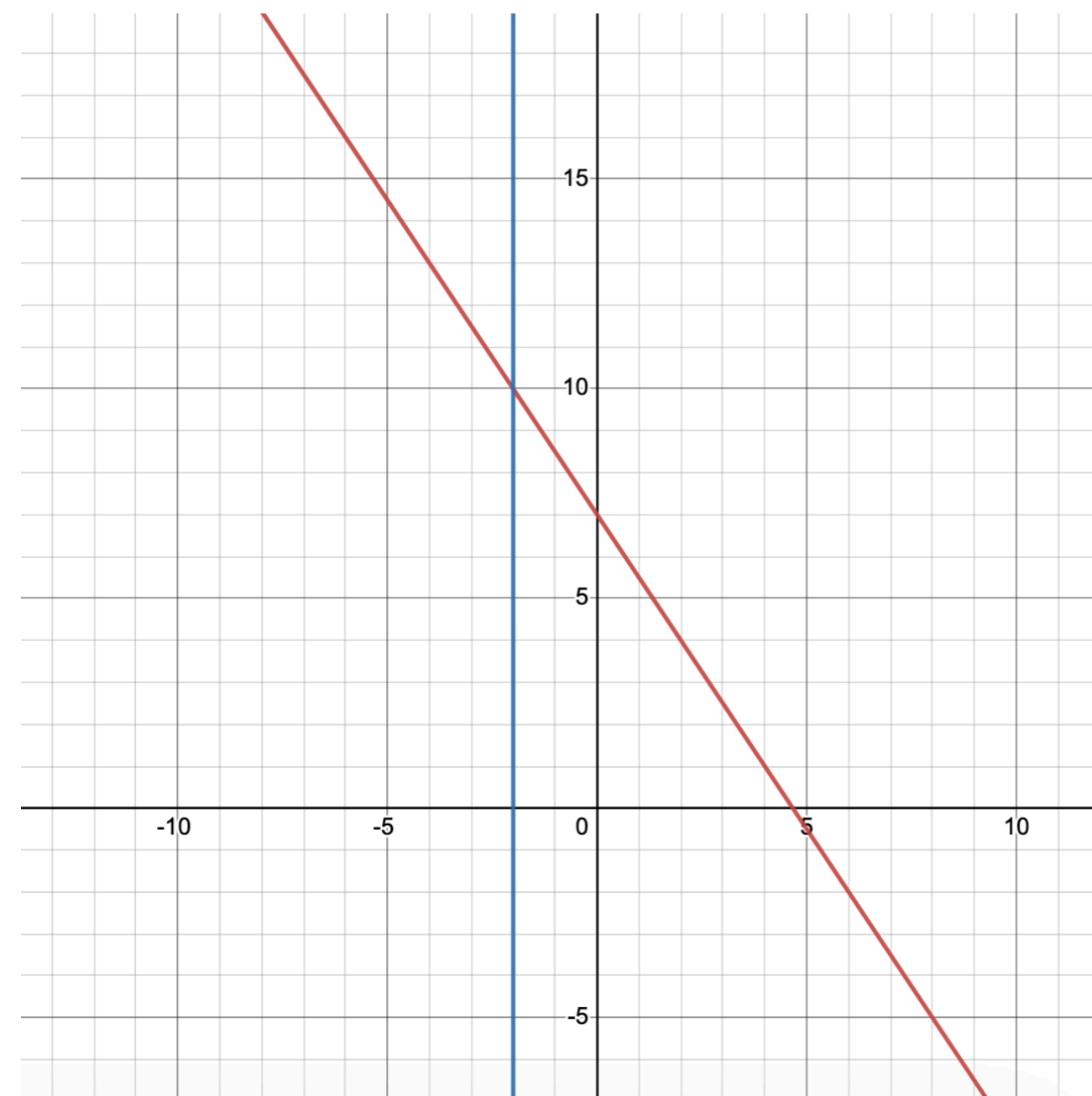
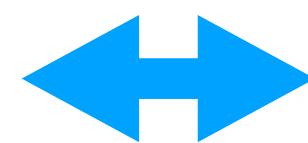
We replace the problem with a new problem
that has the same answer.

$$3x + 2y = 14$$

$$7x + 4y = 26$$



$$R2 - 2R1$$



$$3x + 2y = 14$$

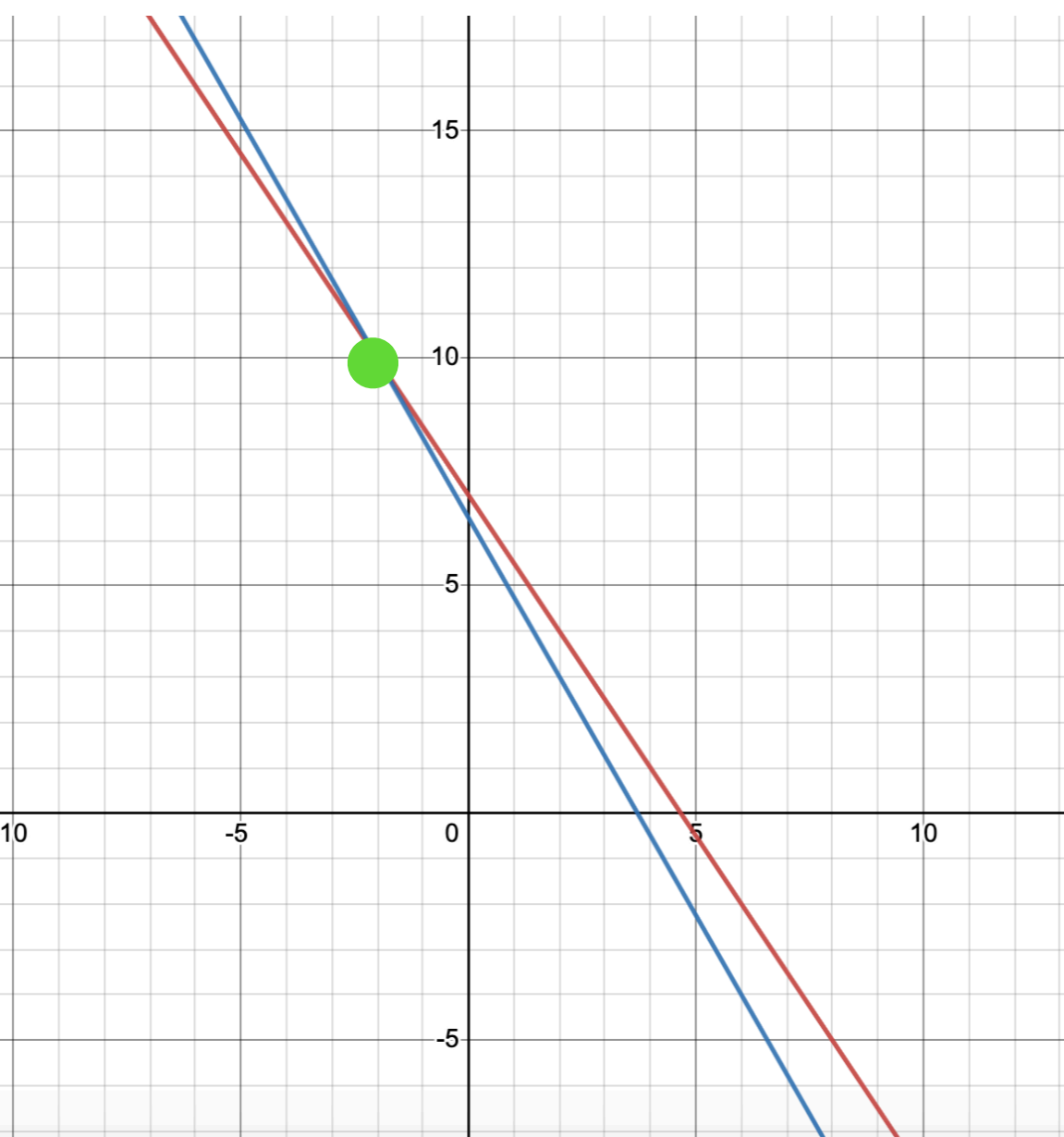
$$x = -2$$

$$x = -2$$

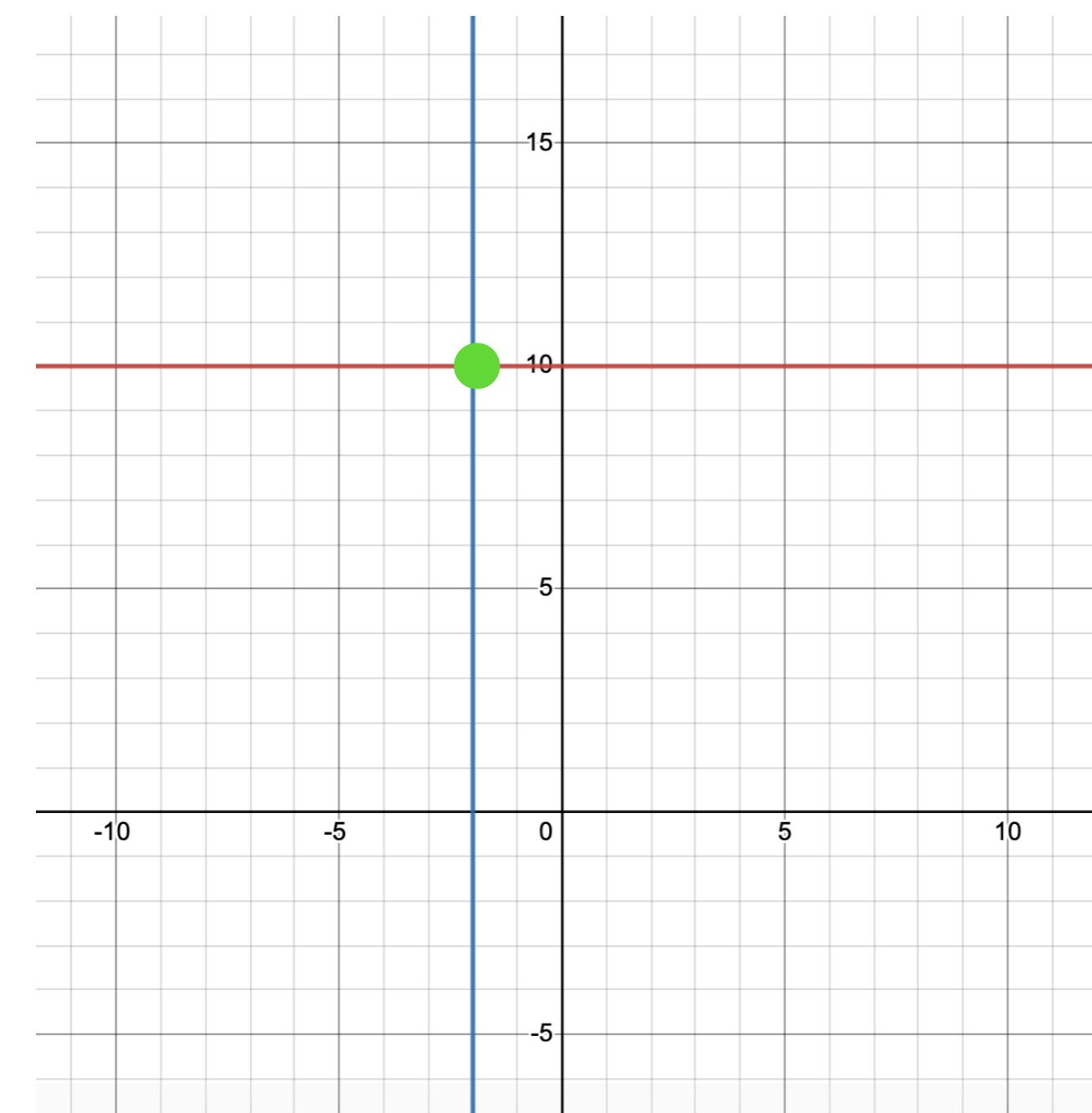
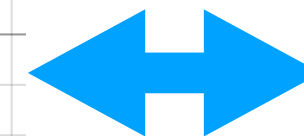
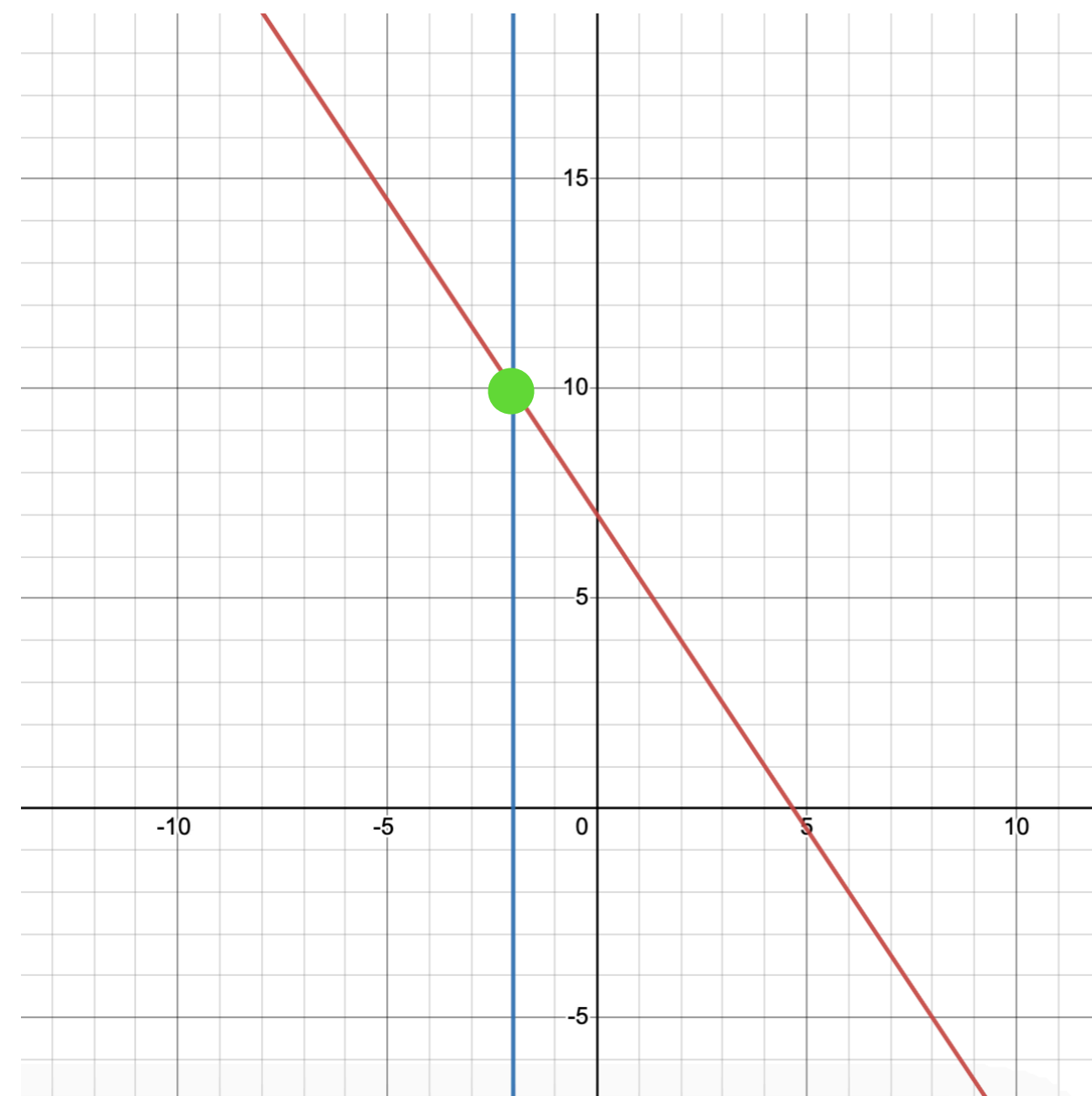
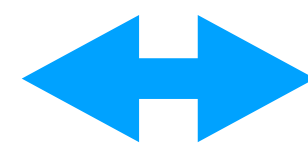
$$y = 10$$

$$3x + 2y = 14$$

$$7x + 4y = 26$$



$$R2 - 2R1$$



$$3x + 2y = 14$$

$$x = -2$$

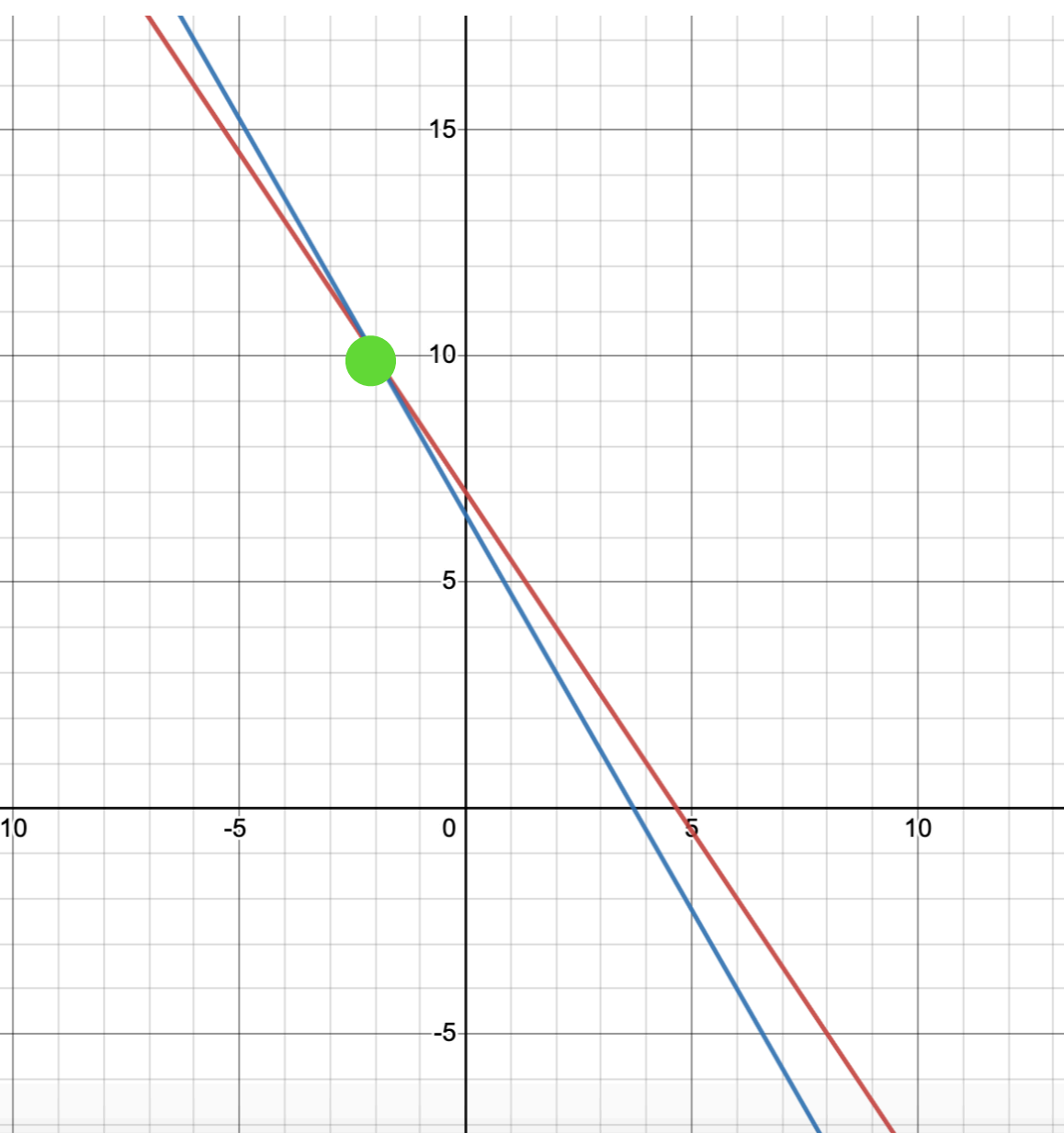
$$x = -2$$

$$y = 10$$

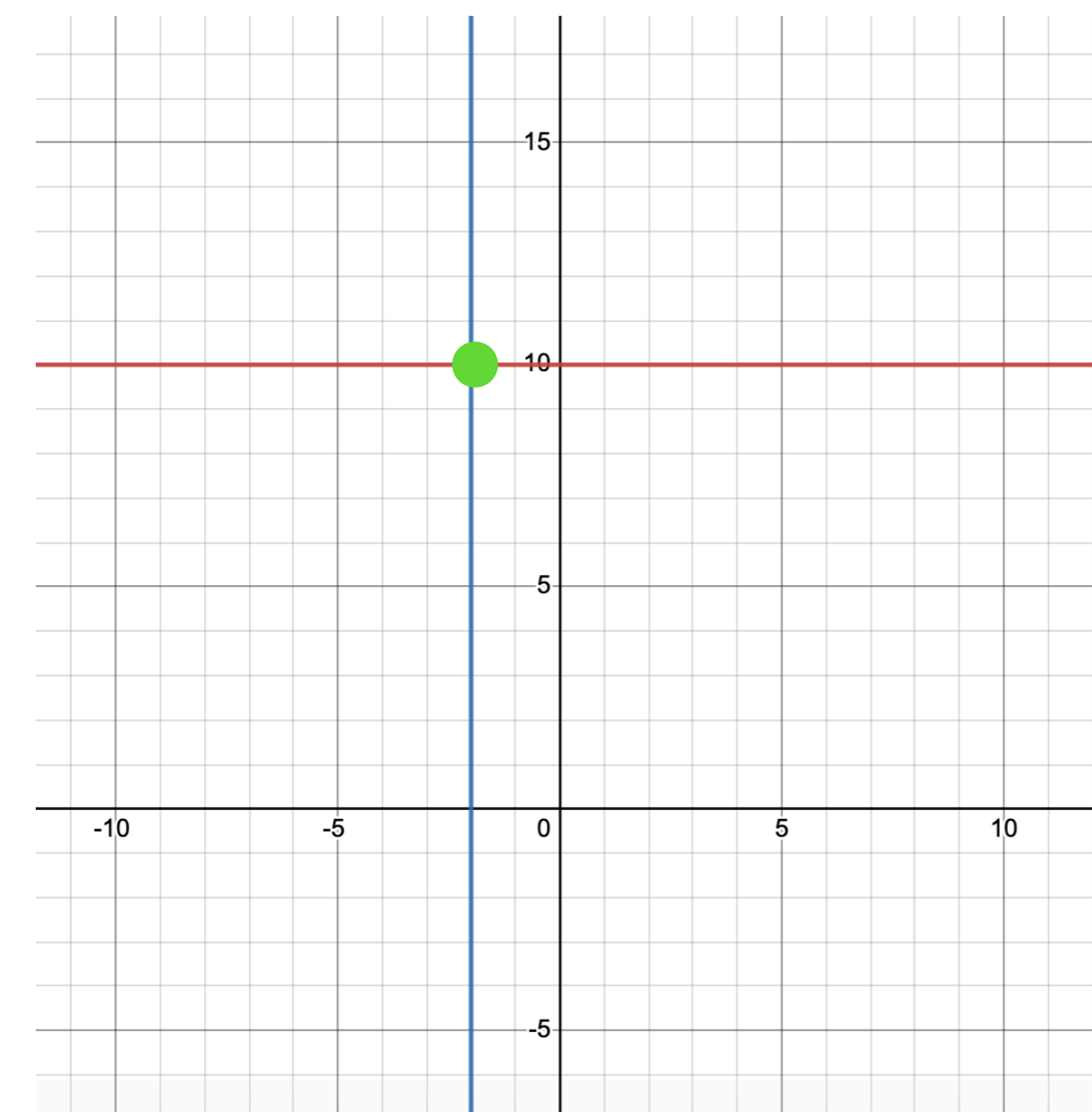
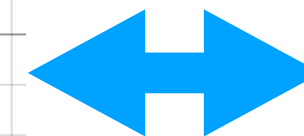
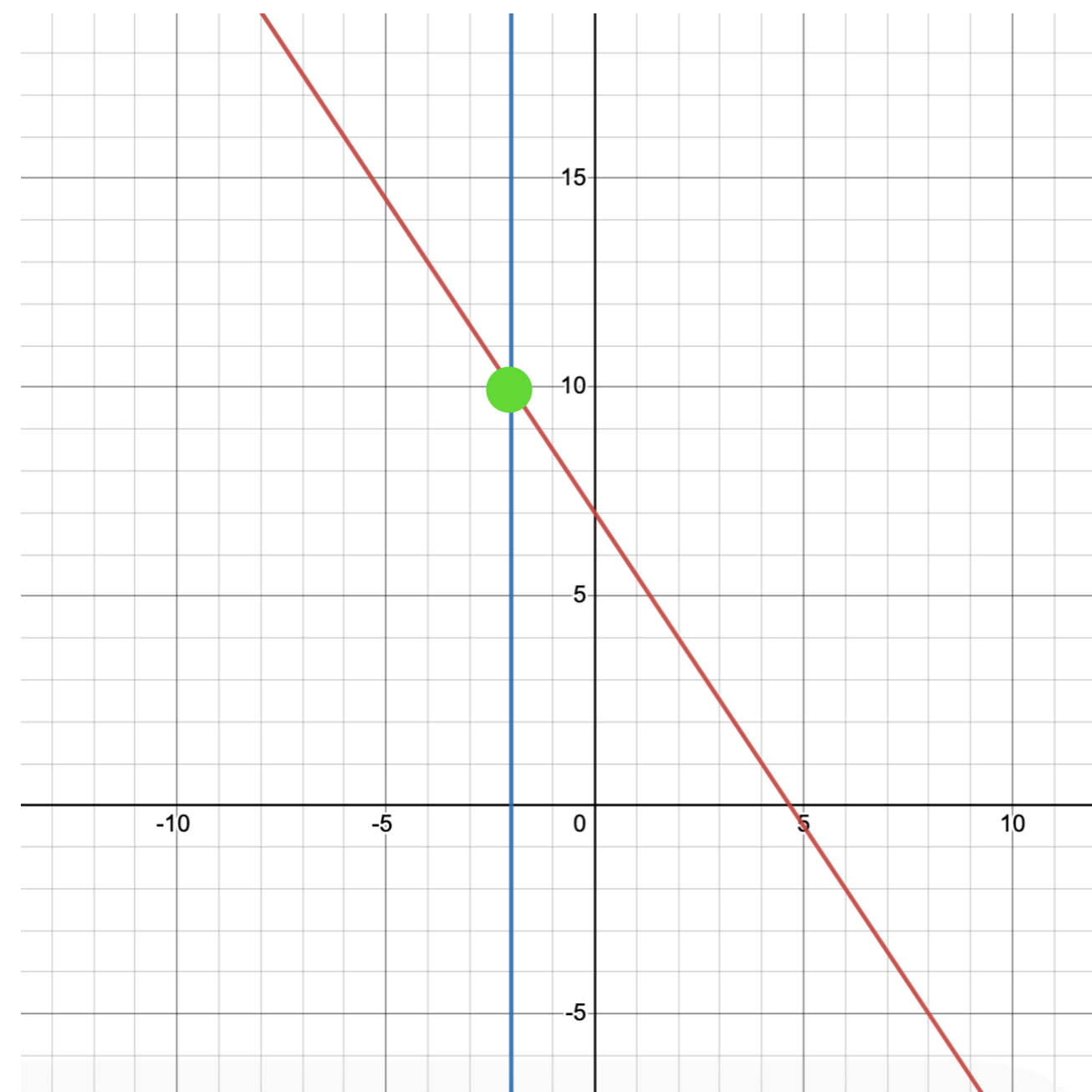
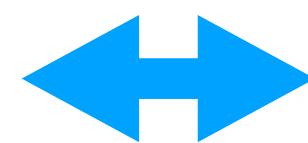
“pivot”

$$3x + 2y = 14$$

$$7x + 4y = 26$$



$$R2 - 2R1$$



$$x = -2$$

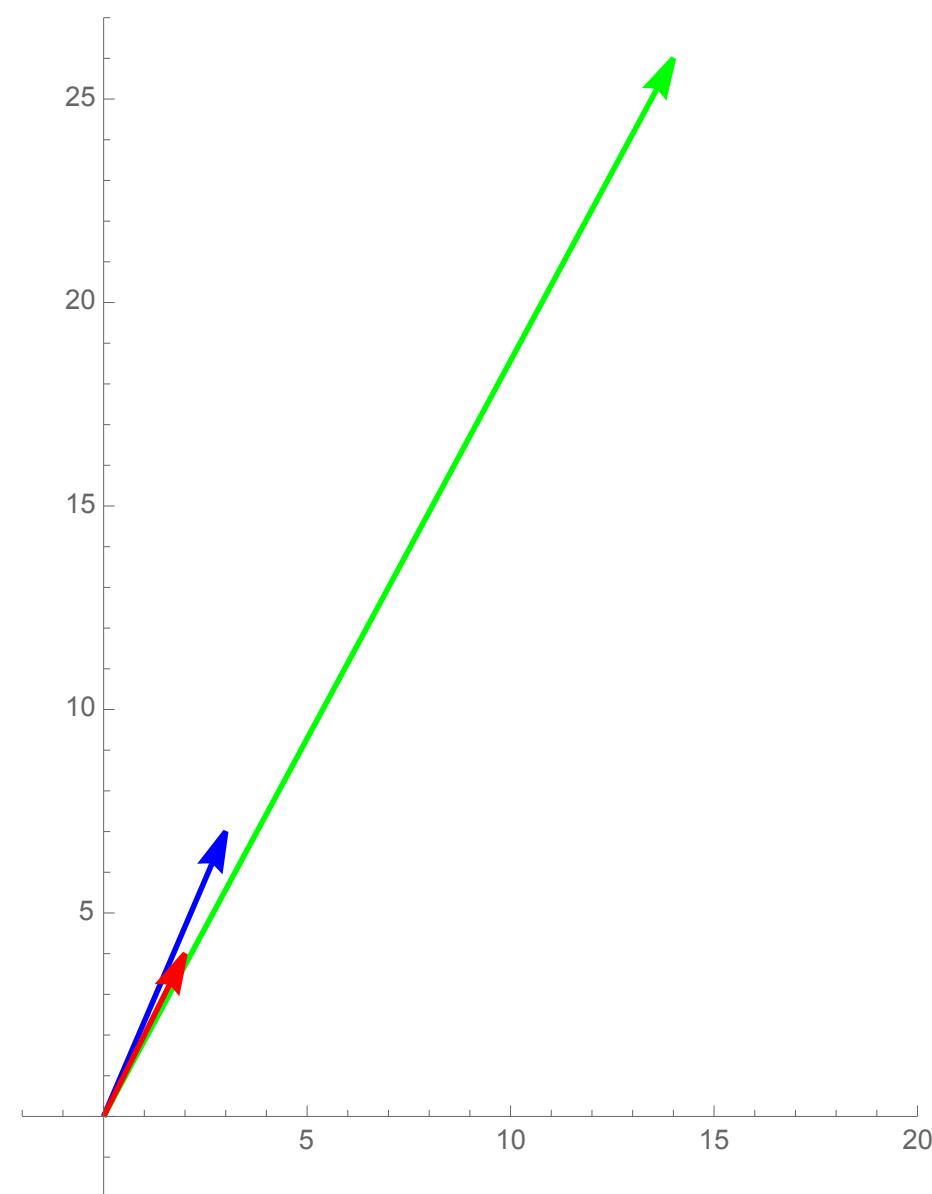
$$y = 10$$

Moves (row operations):

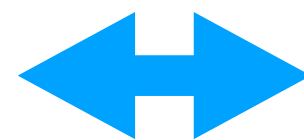
1. swap two rows
2. multiply a row by a non-zero scalar
3. add a multiple of a row to a different row

“pivot”

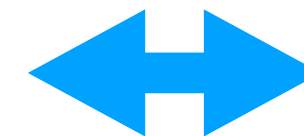
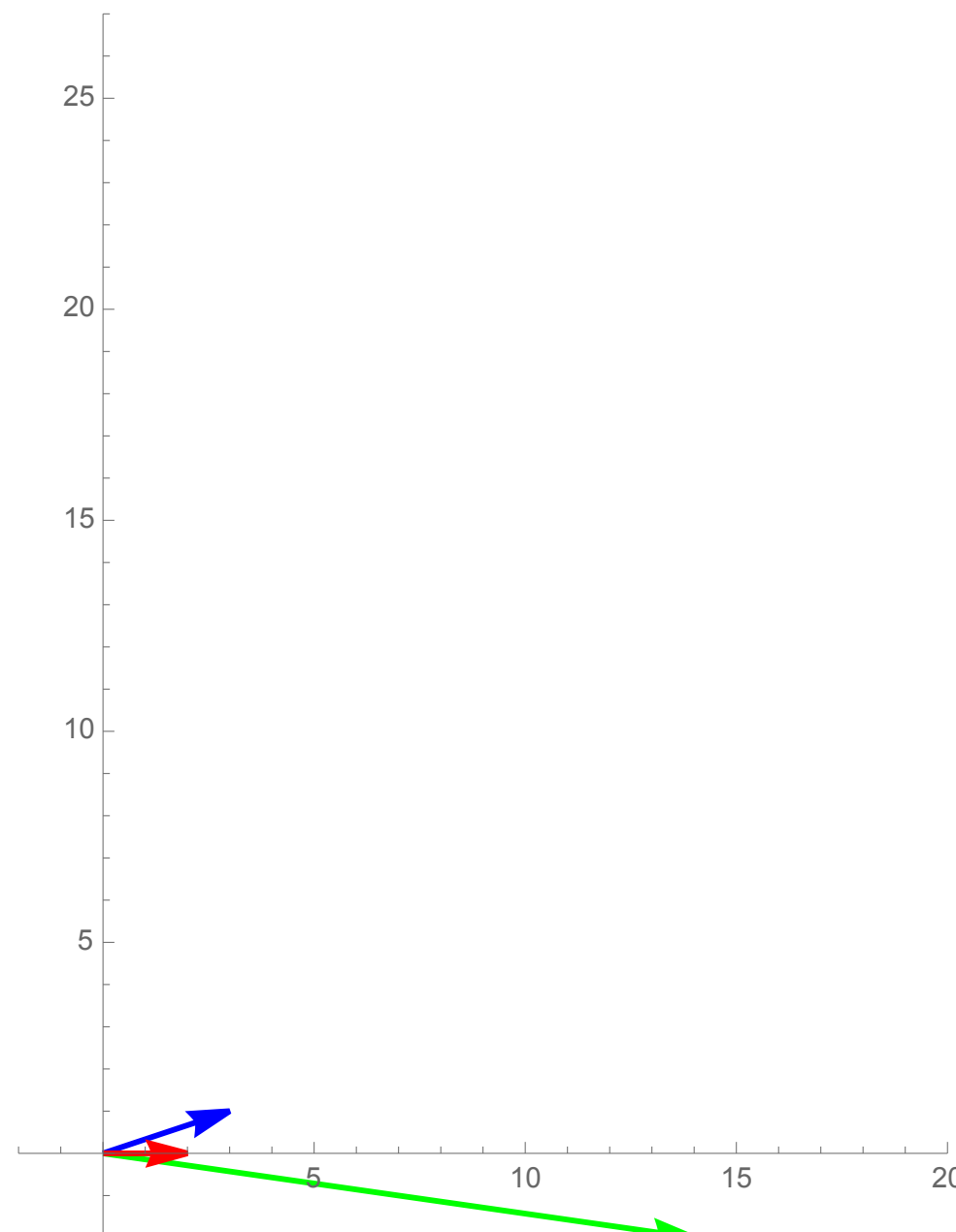
$$x \begin{pmatrix} 3 \\ 7 \end{pmatrix} + y \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 14 \\ 26 \end{pmatrix}$$



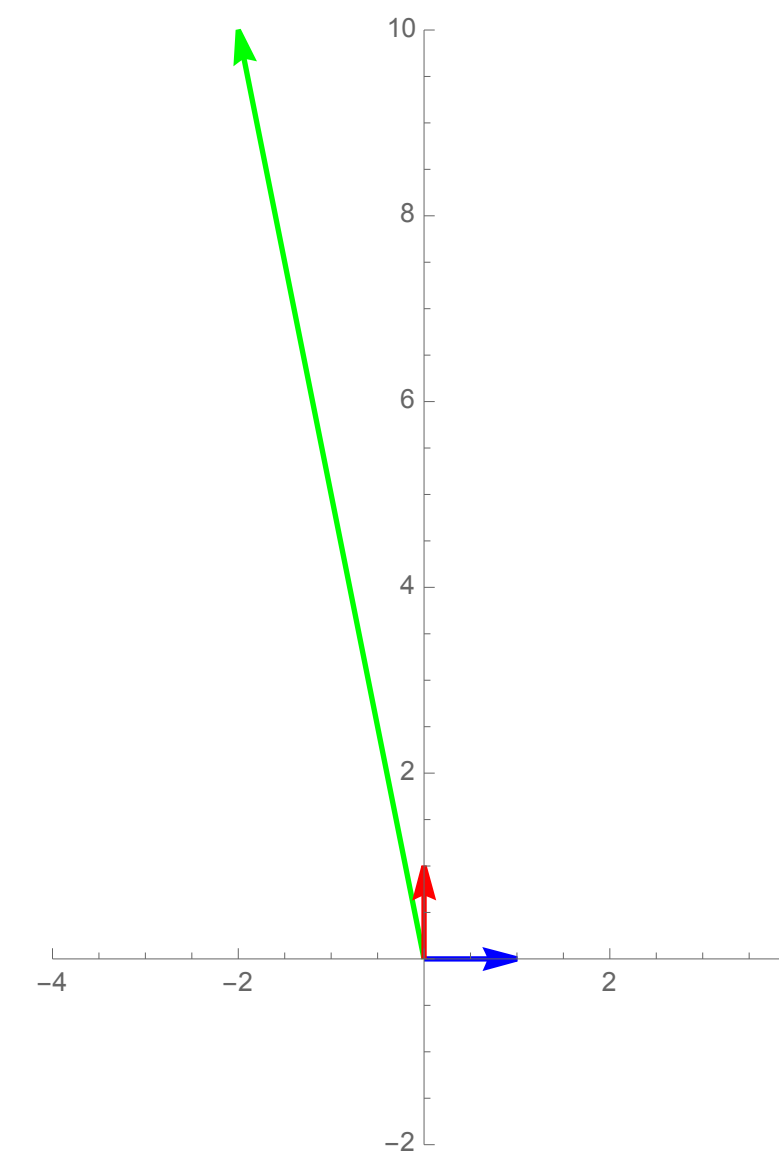
$$R2 - 2R1$$



$$x \begin{pmatrix} 3 \\ 1 \end{pmatrix} + y \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 14 \\ -2 \end{pmatrix}$$



$$x \begin{pmatrix} 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 10 \end{pmatrix}$$

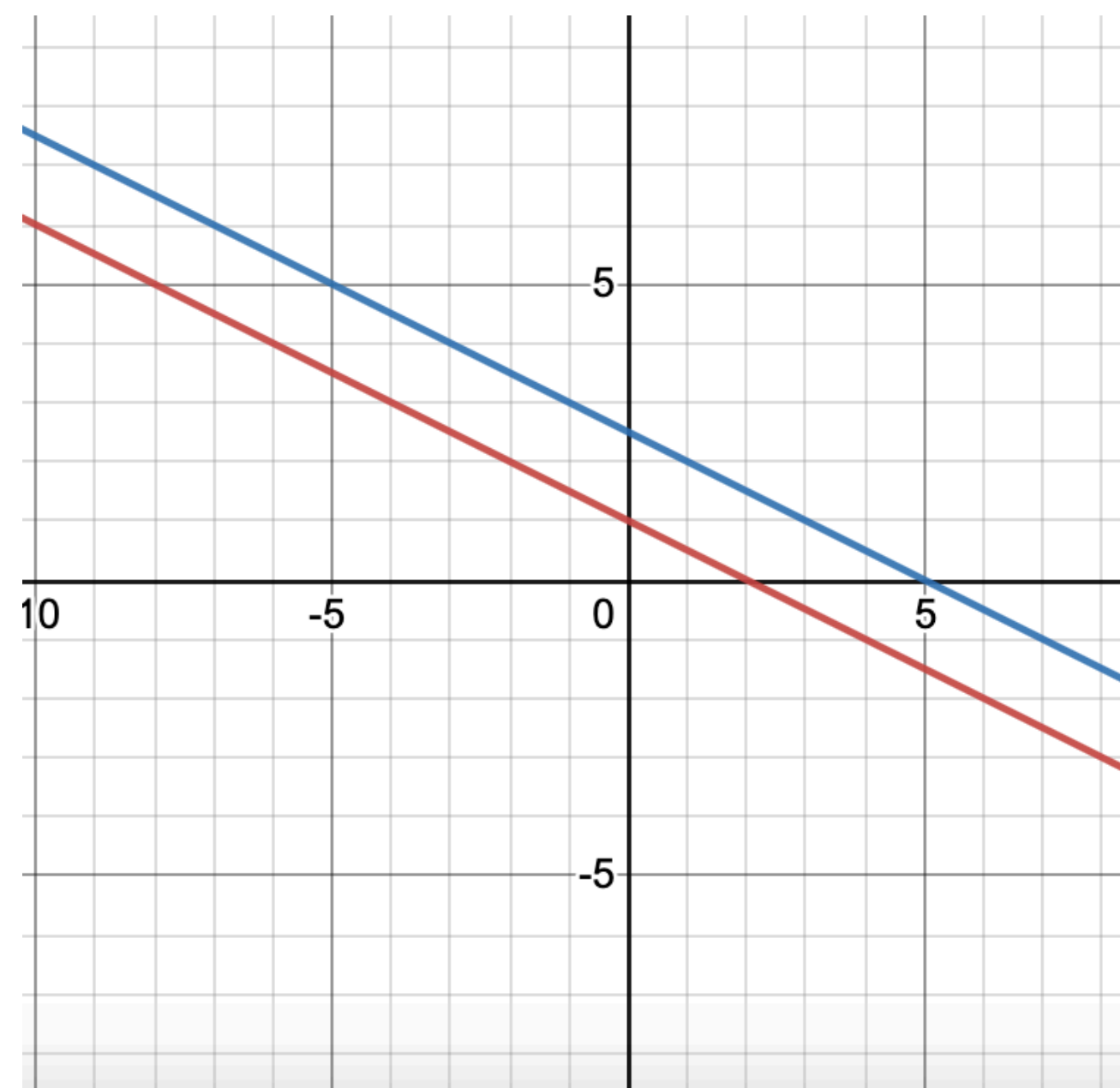


2.3

system of
linear equations

$$x + 2y = 2$$

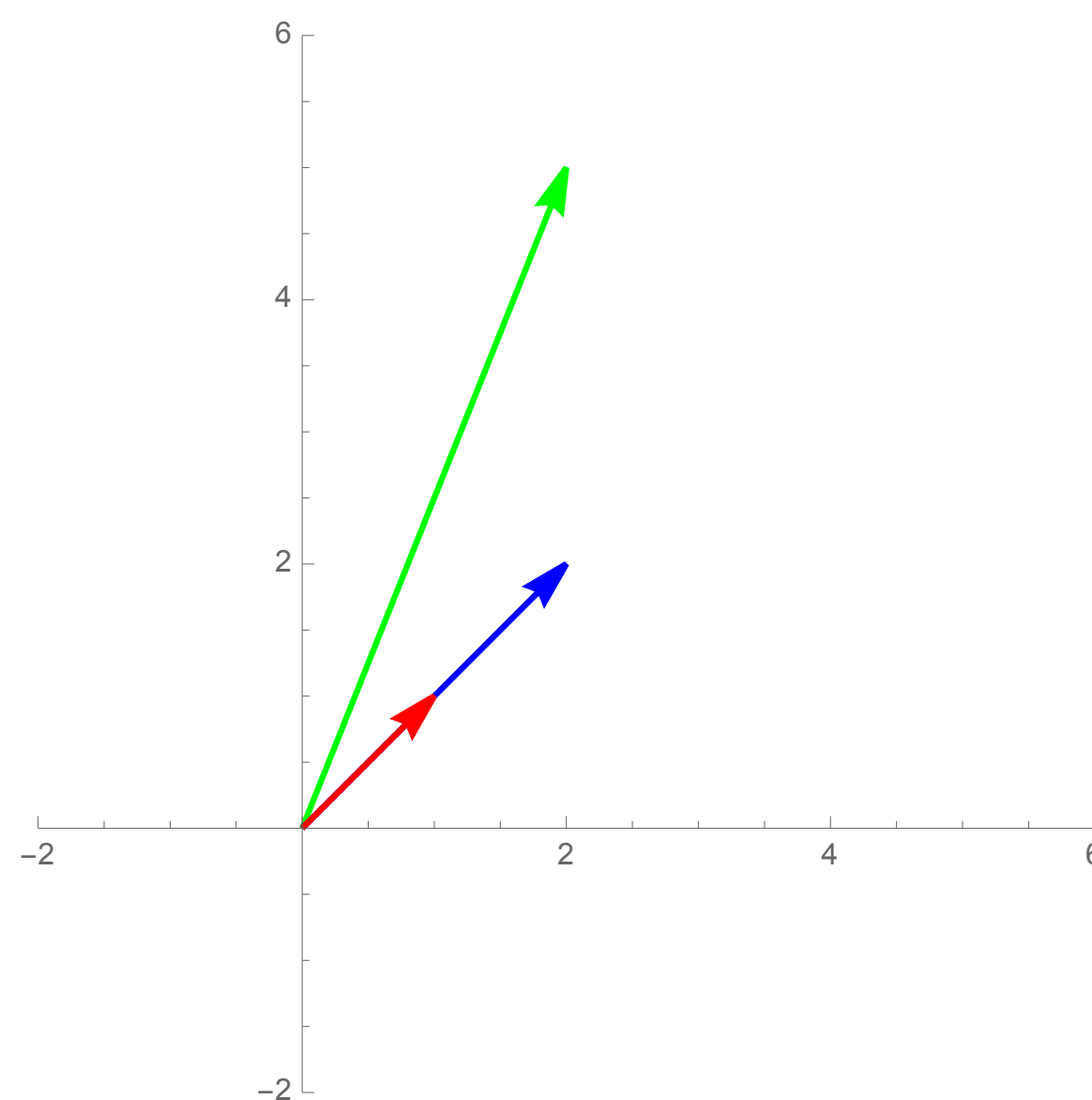
$$x + 2y = 5$$



inhomogeneous systems can fail to have solutions

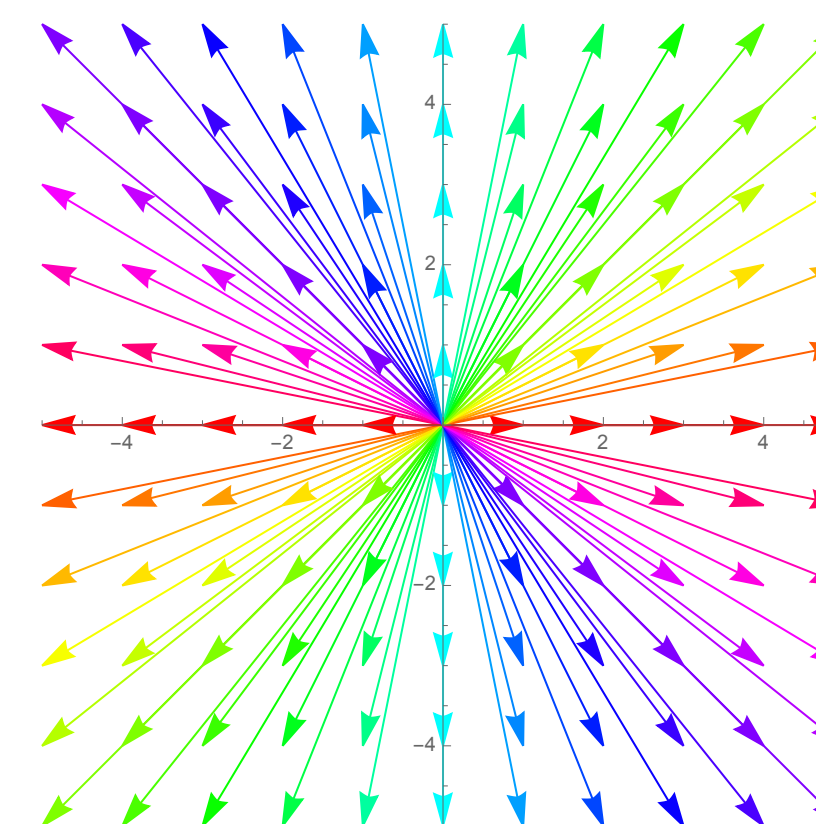
vector equation

$$x \begin{pmatrix} 1 \\ 1 \end{pmatrix} + y \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

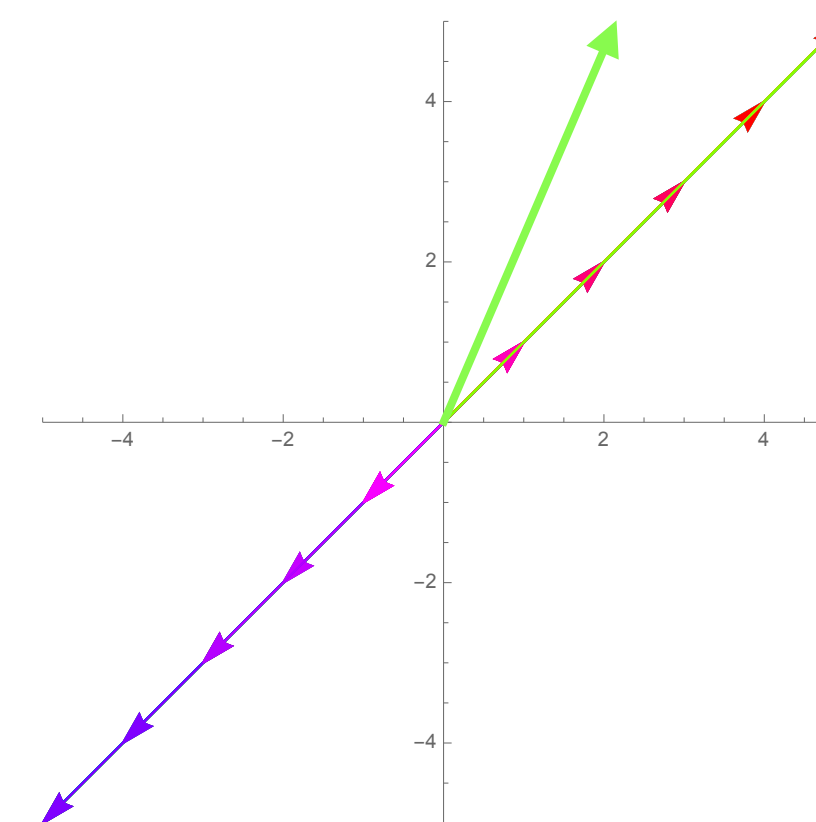


matrix equation

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$



$$\downarrow \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$$

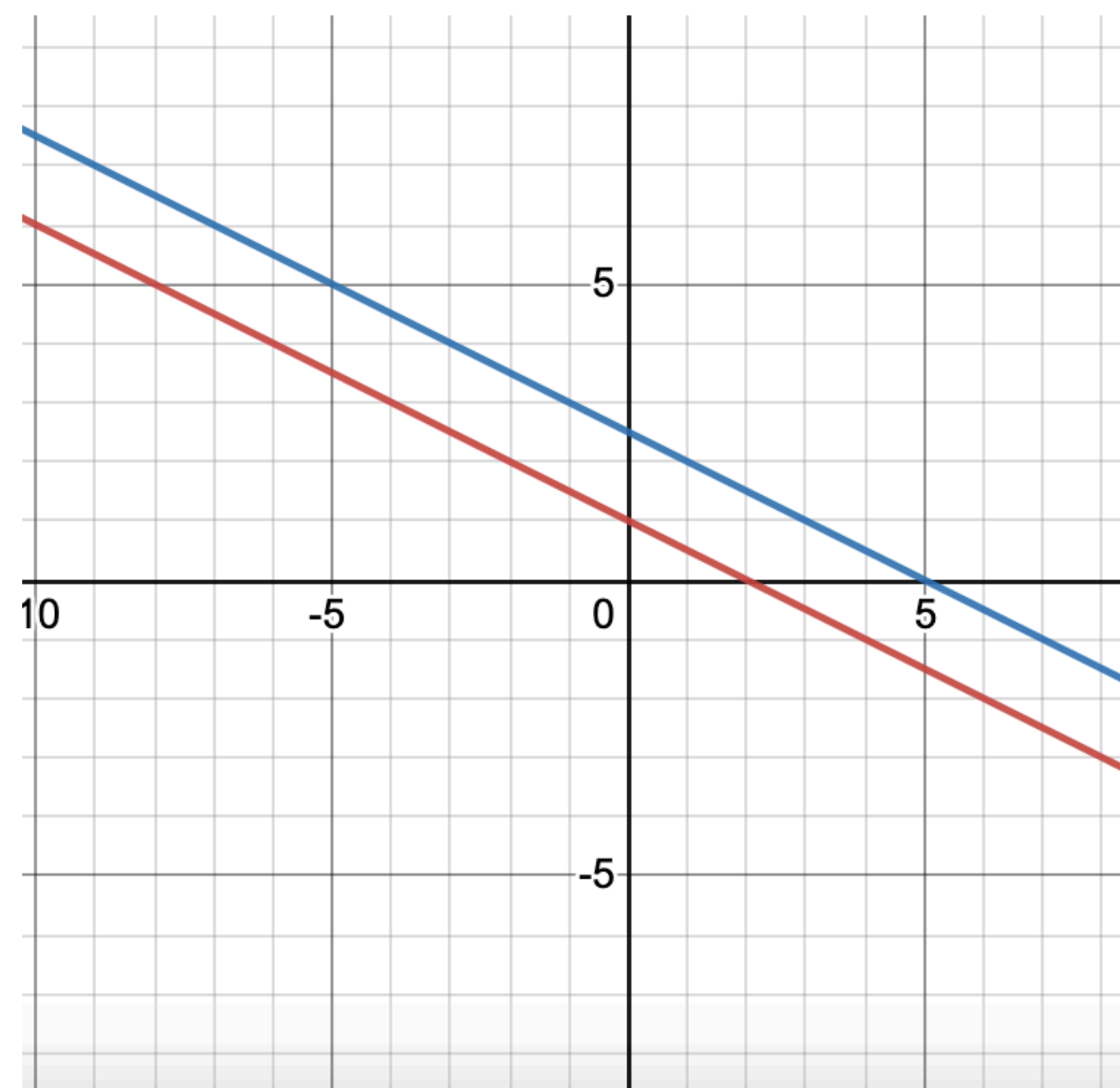


2.3

system of
linear equations

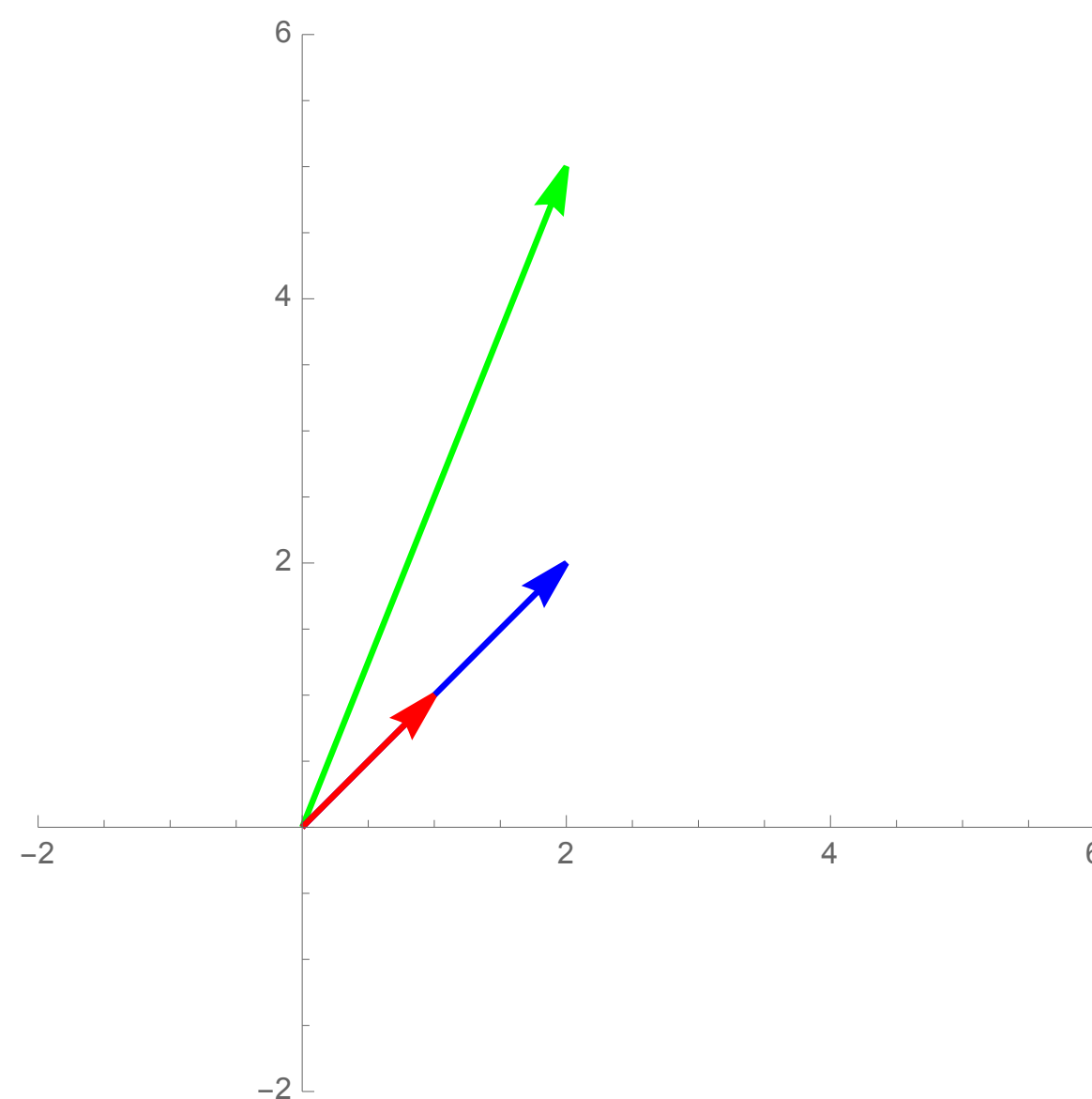
$$x + 2y = 2$$

$$x + 2y = 5$$



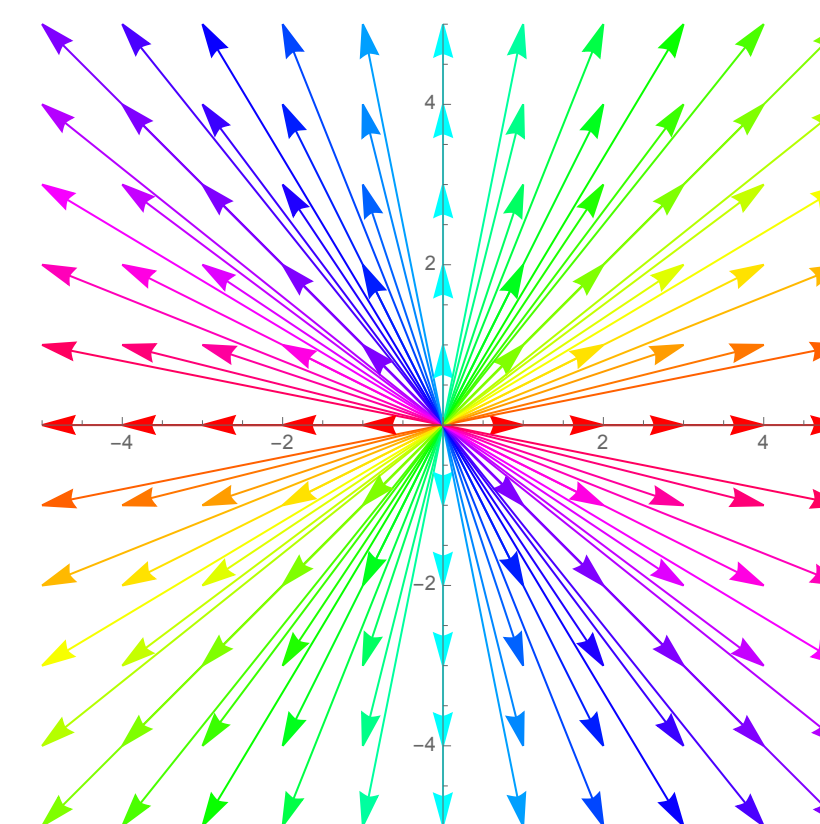
vector equation

$$x \begin{pmatrix} 1 \\ 1 \end{pmatrix} + y \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

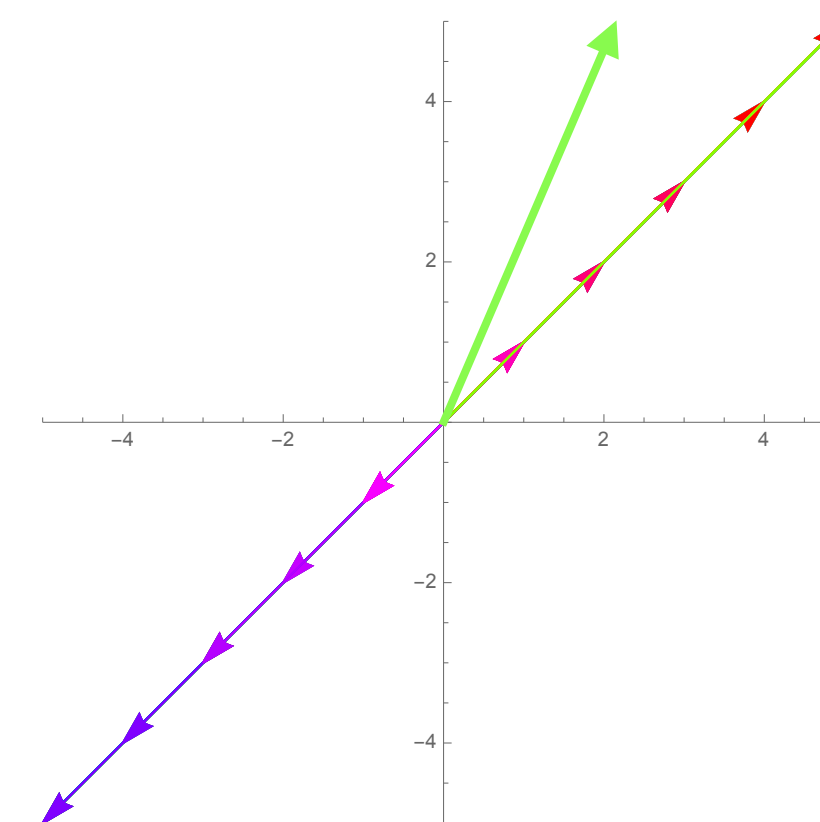


matrix equation

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$



$$\downarrow \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$$



inhomogeneous systems can fail to have solutions

$$\left(\begin{array}{cc|c} 1 & 2 & 2 \\ 0 & 0 & 3 \end{array} \right)$$

$$\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix}$$

I'm taking linear
combinations of your
columns!

$$\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix}$$

I'm taking linear combinations of your columns!

$$\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \left(5 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad 0 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right)$$

$$= \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix}$$

I'm taking linear combinations of your columns!

$$\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \left(5 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad 0 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right)$$

$$= \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} = \begin{pmatrix} 10 & 3 \\ 0 & 1 \end{pmatrix}$$

No, I'm taking linear combinations of your rows!

I'm taking linear combinations of your columns!

$$\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \left(5 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 3 \\ 1 \end{pmatrix} \quad 0 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right)$$

$$= \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix}$$

$$= \begin{pmatrix} 10 & 3 \\ 0 & 1 \end{pmatrix}$$

No, I'm taking linear combinations of your rows!

I'm taking linear combinations of your columns!

$$\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{aligned} \begin{pmatrix} 2(5 & 0) + 3(0 & 1) \\ 0(5 & 0) + 1(0 & 1) \end{pmatrix} &= \begin{pmatrix} 5 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 3 \\ 1 \end{pmatrix} & 0 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} \end{pmatrix} \\ &= \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} = \begin{pmatrix} 10 & 3 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

No, I'm taking linear combinations of your rows!

I'm taking linear combinations of your columns!

$$\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{aligned} \begin{pmatrix} 2(5 & 0) + 3(0 & 1) \\ 0(5 & 0) + 1(0 & 1) \end{pmatrix} &= \begin{pmatrix} 5 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 3 \\ 1 \end{pmatrix} & 0 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} \end{pmatrix} \\ \begin{pmatrix} 10 & 3 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} = \begin{pmatrix} 10 & 3 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

No, I'm taking linear combinations of your rows!

Both are right!

I'm taking linear combinations of your columns!

$$\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{aligned} \begin{pmatrix} 2(5 & 0) + 3(0 & 1) \\ 0(5 & 0) + 1(0 & 1) \end{pmatrix} &= \begin{pmatrix} 5 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 3 \\ 1 \end{pmatrix} & 0 \begin{pmatrix} 2 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} \end{pmatrix} \\ \begin{pmatrix} 10 & 3 \\ 0 & 1 \end{pmatrix} &= \begin{pmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{pmatrix} = \begin{pmatrix} 10 & 3 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

$$\begin{pmatrix} 3 & 4 & 1 \\ 2 & -1 & 6 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 15 & 4 & 1 \\ 10 & -1 & 6 \\ 5 & 4 & 2 \end{pmatrix}$$

acting on the right: scales the first **column** by 5

$$\begin{pmatrix} 3 & 4 & 1 \\ 2 & -1 & 6 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 15 & 4 & 1 \\ 10 & -1 & 6 \\ 5 & 4 & 2 \end{pmatrix}$$

acting on the right: scales the first **column** by 5

$$\begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 4 & 1 \\ 2 & -1 & 6 \\ 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 15 & 20 & 5 \\ 2 & -1 & 6 \\ 1 & 4 & 2 \end{pmatrix}$$

acting on the left: scales the first **row** by 5

$$\begin{pmatrix} 3 & 4 & 1 \\ 2 & -1 & 6 \\ 1 & 4 & 2 \end{pmatrix} \begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 15 & 4 & 1 \\ 10 & -1 & 6 \\ 5 & 4 & 2 \end{pmatrix}$$

acting on the right: scales the first **column** by 5

$$\begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & 4 & 1 \\ 2 & -1 & 6 \\ 1 & 4 & 2 \end{pmatrix} = \begin{pmatrix} 15 & 20 & 5 \\ 2 & -1 & 6 \\ 1 & 4 & 2 \end{pmatrix}$$

acting on the left: scales the first **row** by 5

Another way to see
that matrix multiplication
is in general not commutative!

row reduction moves

multiplication on left by elementary matrices
(square, invertible)

1. swap two rows

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

2. multiply a row by a non-zero scalar

$$\begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}$$

3. add a multiple of a row to a different row

$$\begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$$

(These are examples, not the general form.)

2.2 On left: linear combination of rows. On right: linear combination of columns.

multiplication on left by elementary matrices to achieve row reduction

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 8 \end{pmatrix} \xleftrightarrow[\frac{1}{4}R2]{} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix} \xleftrightarrow{R1 - 2R2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}$$

multiplication on left by elementary matrices to achieve row reduction

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 8 \end{pmatrix} \xleftrightarrow[\frac{1}{4}R2]{\text{blue}} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix} \xleftrightarrow[\text{blue}]{R1 - 2R2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix}$$

multiplication on left by elementary matrices to achieve row reduction

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 8 \end{pmatrix} \xleftrightarrow[\frac{1}{4}R2]{\text{blue double arrow}} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix} \xleftrightarrow[\text{blue double arrow}]{R1 - 2R2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}$$

Compute the inverse matrix for $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$

Three methods for computing matrix inverses (when they exist):

- use geometric intuition
- use row reduction
- use a formula

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array} \right] \Leftrightarrow \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & \frac{ad-bc}{a} & -\frac{c}{a} & 1 \end{array} \right]$$

At least one of a, c is non-zero;
otherwise, matrix is not invertible.
WLOG, assume a is non-zero.

A formula for the inverse of an invertible 2x2 matrix

$$\begin{aligned} \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array} \right] &\Leftrightarrow \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & \frac{ad-bc}{a} & -\frac{c}{a} & 1 \end{array} \right] \Leftrightarrow \left[\begin{array}{cc|cc} a & b & 1 & 0 \\ 0 & 1 & -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{array} \right] \\ &\Leftrightarrow \left[\begin{array}{cc|cc} a & 0 & \frac{ad}{ad-bc} & \frac{-ba}{ad-bc} \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array} \right] \Leftrightarrow \left[\begin{array}{cc|cc} 1 & 0 & \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ 0 & 1 & \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{array} \right] \end{aligned}$$

$ad - bc$ is non-zero; **determinant!**
otherwise, matrix is not invertible.

In particular,
at least one of a, c is non-zero;
WLOG, assume a is non-zero.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

pivot in every column

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

pivot in every column & row

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

pivot in every row

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

pivot in every column

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

linearly independent columns

pivot in every column & row

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

basis columns

pivot in every row

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

spanning columns

pivot in every column

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

linearly independent columns

I-to-I

pivot in every column & row

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

basis columns

I-to-I & onto

pivot in every row

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

spanning columns

onto

pivot in every column

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

linearly independent columns

I-to-I

unique solution to $Ax=b$
(if it exists, depends on b)

pivot in every column & row

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

basis columns

I-to-I & onto

unique solution to $Ax=b$

pivot in every row

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

spanning columns

onto

solution to $Ax=b$ exists

pivot in every column

pivot in every column & row

pivot in every row

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

linearly independent columns

basis columns

spanning columns

I-to-I

I-to-I & onto

onto

unique solution to $Ax=b$
(if it exists, depends on b)

unique solution to $Ax=b$

solution to $Ax=b$ exists

neither $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

pivot in every column

pivot in every column & row

pivot in every row

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

linearly independent columns

basis columns

spanning columns

I-to-I

I-to-I & onto

onto

unique solution to $Ax=b$
(if it exists, depends on b)

unique solution to $Ax=b$

solution to $Ax=b$ exists



neither $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

Next up:

- This week: Chapter 2.4-8 of *LADW*
- ***Change:*** Only Quiz 3 on Friday.
- Open office hours today, 3:30-4:30 in Eckhart 207A.
- Quiz 1 & 2 returned. First requizzing session on Wed (4-5pm Eck 207A)
- Homework 2 is due at the end of the week.
- ***Change:*** I won't be away on Friday, April 21.